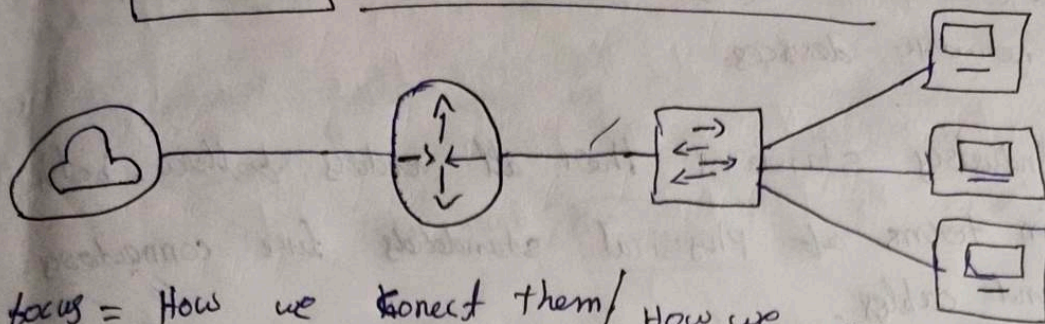


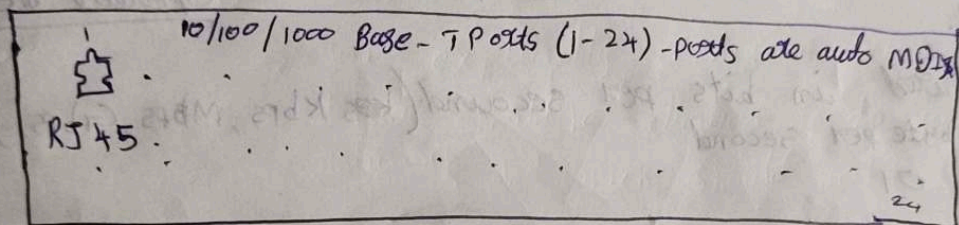
Day-2

Interfaces and cables



focus = How we connect them / How we connect them with cables

switch - they tend to have lots of interfaces to connect end hosts like pc & servers to.



RJ45 (RJ - registered jack)

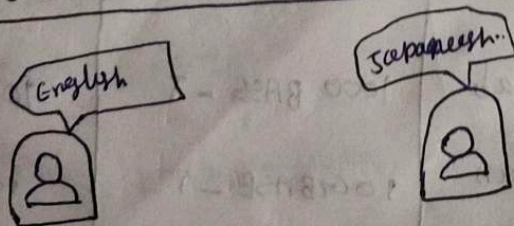
→ RJ-45 connector is used on the end of a copper Ethernet cable

what is Ethernet?

→ Ethernet is a collection of network protocols/standards

→ for the purpose of this lesson, we will focus on types of cabling as defined by Ethernet standards.

why do we need network protocols / standards?



→ there's not going to be a much, or any communication between them.

→ so, they agreed common language between them.

⇒ Network protocols serve that purpose best network devices.

→ Industry standards that all vendors follow, both in terms of physical standards like connectors and cables, as well as logical standards like IP

Bds & Byte

→ connection between devices in network operate at speed.

→ measured in bits per second (Kbps, Mbps, Gbps, ...) not byte per second
bit = 01

byte = 01100111 / 8 bits = 1 byte

Ethernet standards:

⇒ Defined in the IEEE 802.3 standard in 1983.

⇒ IEEE = Institute of Electrical & Electronic Engineers

speed	Common name	IEEE standard	Informal name	maximum Length
10mbps	Ethernet	802.3i	10BASE-T	100m
100Mbps	Fast E.	802.3u	100BASE-T	100m
1Gbps	Gigabit E.	802.3ab	1000BASE-T	100m
10Gbps	10 Ggig E	802.3an	10GBASE-T	100m

BASE = refers to baseband signaling

T = twisted pair (more on that soon!)

UTP cables

→ The copper cables used in Ethernet standards are UTP cables.

Unshielded Twisted Pair

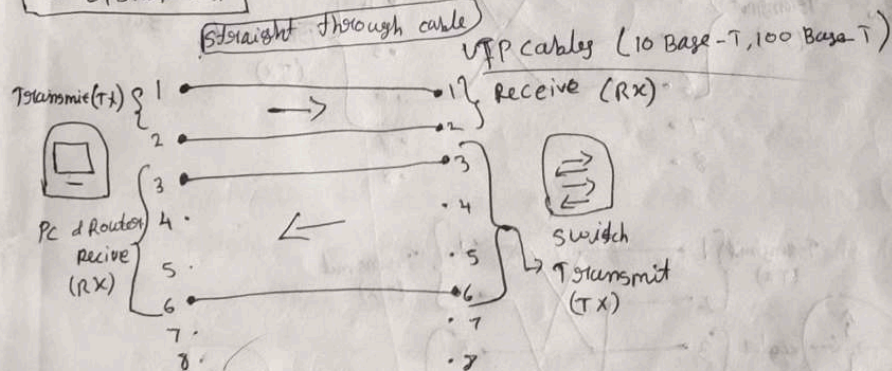
→ don't cover metal ~~shield~~ - vulnerable to electrical interference.

protects against EMI
(Electromagnetic interference)

10 Base-T
100 Base-T = 2 pairs (4 wire)

1000 Base-T
10 GBASE-T = 4 pairs (8 wires)

Straight through cable



→ These numbers represent the pins on the RJ-45

→ The PC ^{or server} transmitting data on pins 1 & 2
Switch receive data on pins 1 & 2

→ Switch transmitting data on pins 3 & 6

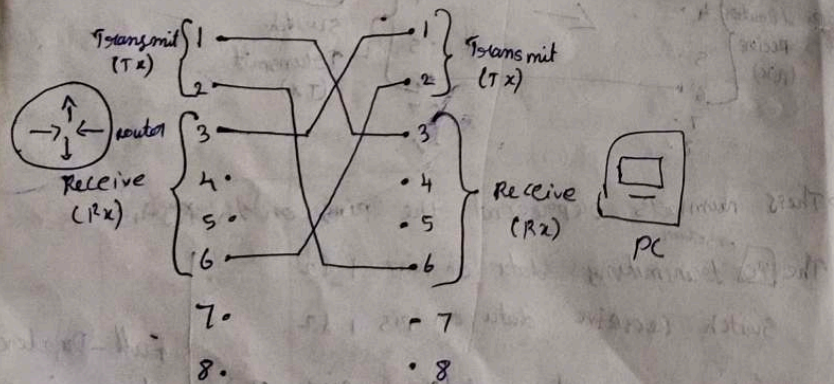
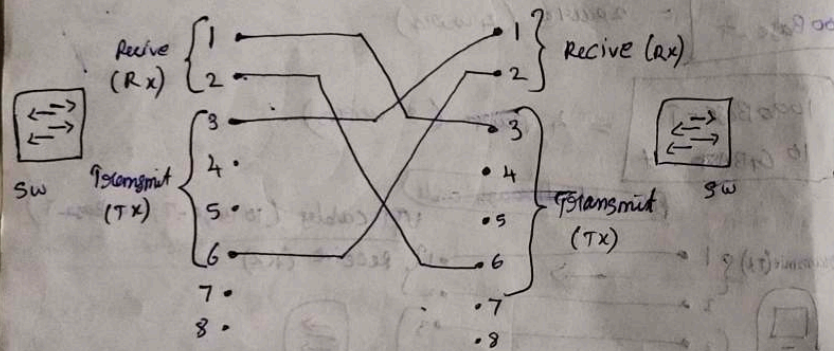
→ PC receive data on pins 3 & 6

Full-Duplex
transmission

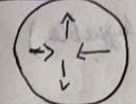
⇒ Full-Duplex transmission means that both devices
 & send data at the same time & no problems like
 collision will occur because they use separate
 wires to transmit and receive data.

How can we successfully connect ~~two~~ two Routers/
 switches
 or
 PC

Crossover cable



10BASE-T, 100BASE-T

Device Type	Transmit (Tx) pins	Receive (Rx) pins
Router 	1 & 2	3 & 6
Firewall 	1 & 2	3 & 6
PC 	1 & 2	3 & 6
Switch 	3 & 6	1 & 2

* modern networking devices have evolved beyond
 having to worry about straight-through or crossover
 cables

modern feature Auto MDI-X

→ Allows devices to automatically detect which pins
 their neighbor is transmitting data on.
 and then adjust which pins they use to
 transmit and receive data.

What about higher speed Copper ethernet cables?

1000BASE-T
 10GBASE-T - each pair is bidirectional meaning

⇒ The last two pairs are 4 & 5, 7 & 8

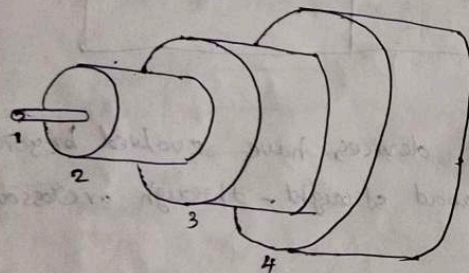
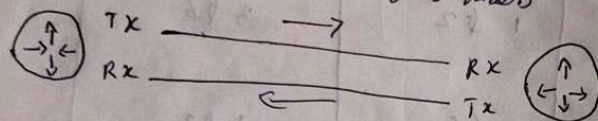
each pair isn't dedicated specifically to
 transmitting or receiving data

Fiber-optic Connections:

→ SFP Transceiver (small form-factor pluggable)

fiber optic cable.

⇒ Rather than an electrical signal over copper wiring these cables send light over the glass fibers



1) the fiber glass core itself
→ Light is transmitted down this core to transmit data from one device to another.

2) Cladding that reflects light

3) a protective buffer

4) the outer jacket of the cable

Types of fiber-optic cables:

* single mode

* multi mode

Multimode Fiber:

- Core diameter is wider than single-mode
- Allows multiple angles (modes) of light waves to enter the fiber glass core

- Allows longer cables than UTP, but shorter cables than single-mode fiber
- Cheaper than single-mode fiber (due to cheaper LED-based SFP transmitters).

Single-mode fiber:

- core diameter is narrower than multimode fiber
- Light enters at a single angle (mode) from a laser-based transmitter.
- Allows longer cables than both UTP & multimode fiber
- more expensive than multimode.

Name	IEEE standard	Speed	Cable Type	Maximum Length
1000BASE-LX	802.3Z	1Gbps	Multimode or single	550m (MM) 5km (SM)
10GBASE-SR	802.3ae	10Gbps	multimode	400m
100GBASE-LR	802.3ae	10Gbps	single mode	10km
10GBASE-ER	802.3ae	10Gbps	SM	30km

UTP

- Lower cost
- Shorter maximum distance than fiber-optic (~100m)
- Can be vulnerable to EMI (Electromagnetic Interference)
- RJ 45 ports used with UTP are cheaper than SFP ports
- Emit (leak) a faint signal outside of the cable, which can be copied (=security risk)

Fiber-optic

- Higher cost
- Longer maximum distance than UTP
- No vulnerability to EMI
- SFP ports are more expensive than RJ 45 ports (single-mode is more expensive)
- Does not emit any signal outside of the cable